UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.
A2	TRUE	A web browser displays hypertext documents.	Browsers render web pages.
A3	TRUE	A higher sampling rate generally produces a more accurate recording.	More samples capture more detail.
A4	TRUE	A graphics tablet is used to draw directly into a computer.	A tablet is a drawing input device.
A5	TRUE	HTML is used to structure content on web pages.	HTML marks up web content.
A6	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A7	TRUE	MPEG is a common family of video compression formats.	MPEG defines video compression standards.
A8	FALSE	Streaming requires the whole file to download before playback.	Streaming can play while downloading.
A9	TRUE	Chroma key compositing replaces a colour background with another image.	Chroma key removes a selected colour.
A10	FALSE	The target audience has no influence on multimedia design.	Design is shaped by the intended audience.
A11	TRUE	The amplitude of a sound wave determines its loudness.	Larger amplitude means louder sound.
A12	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.
A13	TRUE	USB is a plug-and-play connection standard.	USB devices are recognised automatically.
A14	TRUE	A scanner is an input device.	A scanner captures documents into the computer.
A15	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	M	Sampling	converting sound into digital values
B2	B	Browser	software that displays web pages
B3	K	Serif	font with small strokes at character ends
B4	G	MIDI	standard for musical notes and instructions
B5	A	Multimedia	integration of text, graphics, audio, video and animation
B6	N	Bitmap	image made of pixels
B7	L	Keyframe	important frame in an animation
B8	D	Storyboard	visual sequence plan
B9	E	Linear multimedia	presentation without user control of sequence
B10	J	Sans serif	font without small strokes
B11	F	Vector	image made of mathematical objects
B12	O	Scanner	input device that digitises documents
B13	C	Frame rate	frames displayed per second
B14	I	Kerning	adjustment of space between character pairs
B15	H	Hypermedia	linked multimedia elements

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	C	Modelling and rendering
C2	A	Printed output
C3	D	Display web pages
C4	A	Vector
C5	A	Important frames in an animation
C6	C	Automatic device recognition
C7	B	In-between frames
C8	D	Drawing each frame by hand
C9	D	Non-linear
C10	B	Card-based
C11	D	Frames per second
C12	A	Along a predefined route
C13	A	CMYK
C14	C	Multimedia with hyperlinks
C15	C	One shape into another
C16	A	Structure web content
C17	C	Frequency
C18	C	Permission to use a work
C19	B	Number of pixels in an image
C20	B	Drawing input
C21	A	Smooth jagged edges
C22	A	Assemble multimedia elements
C23	A	Messages with a beginning and end
C24	B	PNG
C25	C	Merging of media and computing
C26	D	A simulated environment
C27	C	A visual sequence plan
C28	B	An early model of a product
C29	A	A cinema presentation
C30	A	Loudness

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PART D - WORKED ANSWERS

QUESTION 1

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

QUESTION 2

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(3 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(4 marks)

QUESTION 3

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(2 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(4 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

QUESTION 4

Topic: text

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed

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material.

Sans serif is cleaner at screen sizes; serif aids reading in print.

(2 marks)

b) Describe TWO roles of hypertext in multimedia.

It links related content and supports non-linear navigation.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

Clear headings and links guide users through HTML pages and make the information easier to scan and follow.

(5 marks)

END OF MARKING SCHEME

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